

# Feature Normalization and Regularization

## COSHE Seminar

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## Background

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## Interests

- High-dimensional statistics
- Regularization and sparsity
- Optimization and computational statistics
- Research software, reproducibility, data visualization, and data science in general

## This Talk

I am going to present *The Choice of Normalization Influences Shrinkage in Regularized Regression* (Larsson and Wallin [2025](#)).<sup>1</sup>

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- But there is no research on this!
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## Conclusions

- Normalization has strong effect on shrinkage and feature selection
- In mixed data, choice of normalization implicitly weighs features' importances.



**Figure 1:** Joint work with Jonas Wallin

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Preliminaries and Motivation

A Little Theory

Experiments

## Preliminaries and Motivation

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- Data consists of a **fixed** matrix of features  $\mathbf{X} \in \mathbb{R}^{n \times p}$  and a response vector  $\mathbf{y} \in \mathbb{R}^n$ .
- $\mathbf{y}$  comes from a linear model, that is,

$$y_i = \beta_0^* + \mathbf{x}_i^\top \boldsymbol{\beta}^* + \varepsilon_i \quad \text{for } i \in 1, \dots, n,$$

where  $\boldsymbol{\beta}^*$  is the vector of *true* coefficients.

- $\varepsilon_i$  is the measurement noise, generated from some random variable.

# The Elastic Net

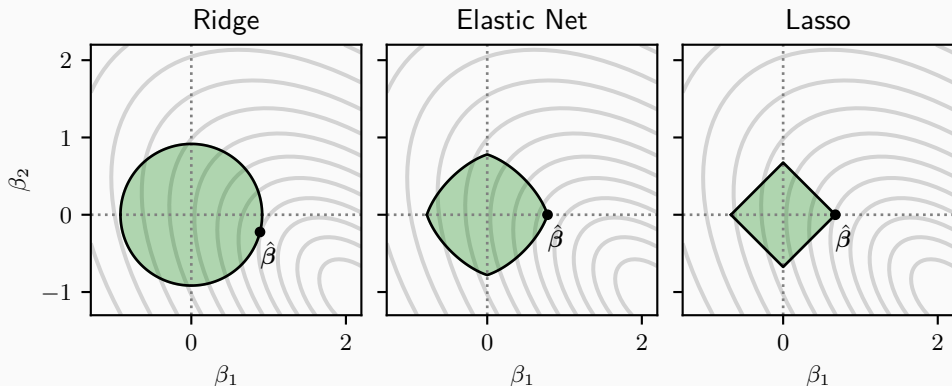
Linear regression plus a combination of the  $\ell_1$  (lasso) and  $\ell_2$  (ridge) penalties:

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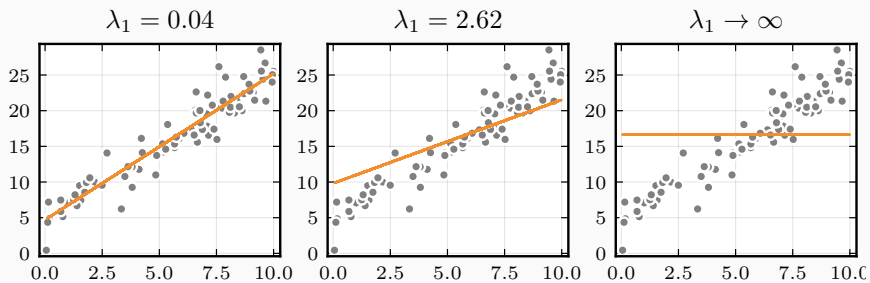
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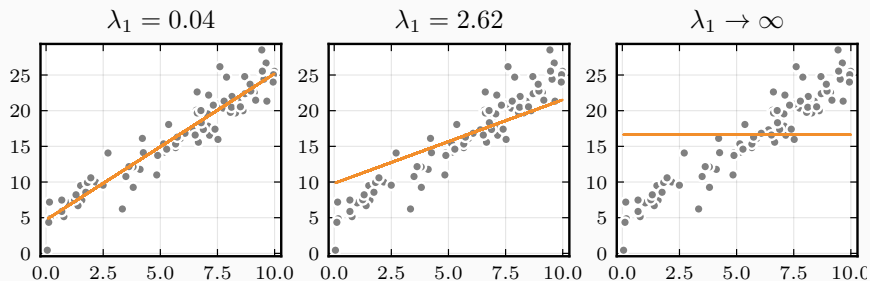


**Figure 2:** Elastic net is a combination of the lasso and ridge.

# Regularization



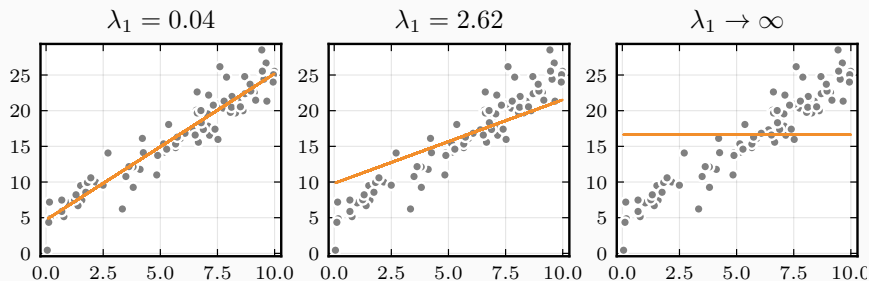
**Figure 3:** Regularization for a simple linear regression problem



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- To overcome overfitting



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- Interpretability
- The sparsity bet

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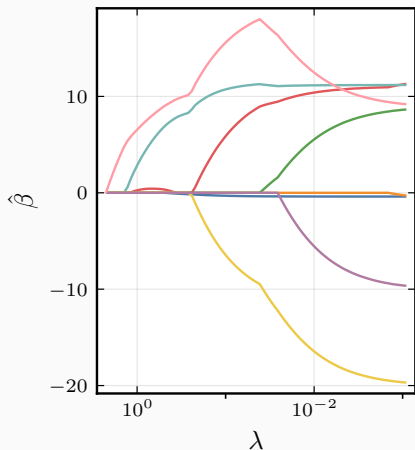
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- For each  $\alpha$ , solve the elastic net over a sequence of  $\lambda$ : the **elastic net path**.



**Figure 4:** The elastic net path

## Sensitivity to Scale

Lasso and ridge penalties are **norms**—feature scales matter!

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### Example

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$$\mathbf{X} \sim \text{Normal} \left( \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \right), \quad \boldsymbol{\beta}^* = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

and set  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}^* + \varepsilon$ , with  $\varepsilon \sim \text{Normal}(0, \sigma_\varepsilon^2)$ .

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Large scale means less penalization because the size of  $\beta_j$  can be smaller for an equivalent effect (on  $\mathbf{y}$ ).

Scale sensitivity can be mitigated by normalizing the features. Let  $\tilde{X}$  be the normalized feature matrix, with elements

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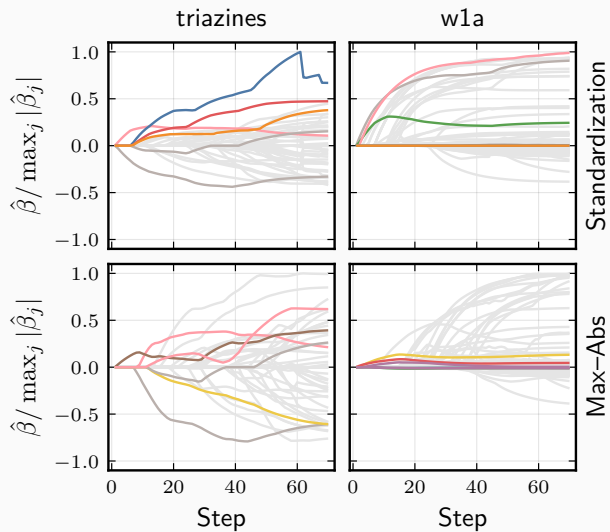
For interpretation, we typically transform the coefficients back to their original scale via

$$\hat{\beta}_j = \frac{\hat{\beta}_j^{(n)}}{s_j} \quad \text{for } j = 1, 2, \dots, p,$$

where  $\hat{\beta}_j^{(n)}$  is a coefficient from the normalized problem.

**Table 1:** Common ways to normalize  $X$

| Normalization   | Centering ( $c_j$ )                           | Scaling ( $s_j$ )                                        |
|-----------------|-----------------------------------------------|----------------------------------------------------------|
| Standardization | $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$ | $\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$ |
| Min–Max         | $\min_i(x_{ij})$                              | $\max_i(x_{ij}) - \min_i(x_{ij})$                        |
| Max–Abs         | 0                                             | $\max_i( x_{ij} )$                                       |
| Adaptive Lasso  | 0                                             | $\beta_j^{\text{OLS}}$                                   |



**Figure 5:** Normalization matters. Lasso paths under two different types of normalization (standardization and max-abs normalization). The union of the first ten features selected in any of the settings are colored.

**Table 2:** Lasso coefficients on test sets, with  $\lambda$  set from 5-fold cross-validation repeated 5 times. We show the five largest coefficients in magnitude for each data set for the standardization setting and corresponding coefficients for the max-abs normalization setting.

| housing                    |                                | triazines                  |                                | w1a                        |                                |
|----------------------------|--------------------------------|----------------------------|--------------------------------|----------------------------|--------------------------------|
| $\hat{\beta}_{\text{std}}$ | $\hat{\beta}_{\text{max-abs}}$ | $\hat{\beta}_{\text{std}}$ | $\hat{\beta}_{\text{max-abs}}$ | $\hat{\beta}_{\text{std}}$ | $\hat{\beta}_{\text{max-abs}}$ |
| -0.63                      | -0.68                          | 0.17                       | 0.0                            | 1.8                        | 0.0                            |
| -1.4                       | -0.78                          | 0.069                      | 0.0                            | 1.8                        | 0.78                           |
| 0.27                       | 0.0                            | 0.028                      | 0.0                            | 1.8                        | 0.63                           |
| -0.99                      | -0.34                          | 0.071                      | 0.0                            | 1.4                        | 0.080                          |
| 2.8                        | 3.1                            | 0.029                      | 0.0                            | 1.7                        | 0.0                            |

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For the elastic net, you should scale the penalty weights—not the features.

When mixing binary and normal features, normalization implicitly weighs their importance.

## A Little Theory

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## Orthogonal Features

There is no explicit solution to the elastic net problem in general.

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<sup>2</sup>We have also assumed that the features are mean-centered here.

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But if we assume that the features are orthogonal, that is

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then there is:<sup>2</sup>:

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}_j^\top \mathbf{y})}{s_j(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)},$$

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$$S_\lambda(z) = \text{sign}(z) \max(|z| - \lambda, 0).$$

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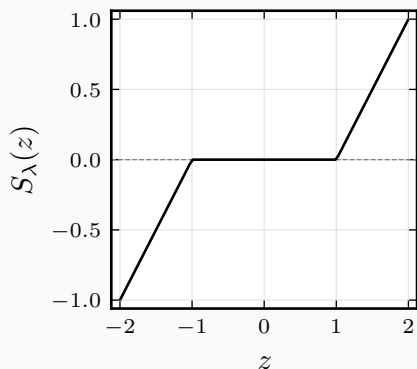
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**Figure 6:** Soft thresholding

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In the noiseless case, we have

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$s_j = q - q^2$  for lasso and  $s_j = \sqrt{q - q^2}$  for ridge removes effect of  $q$ , which suggests the parametrization

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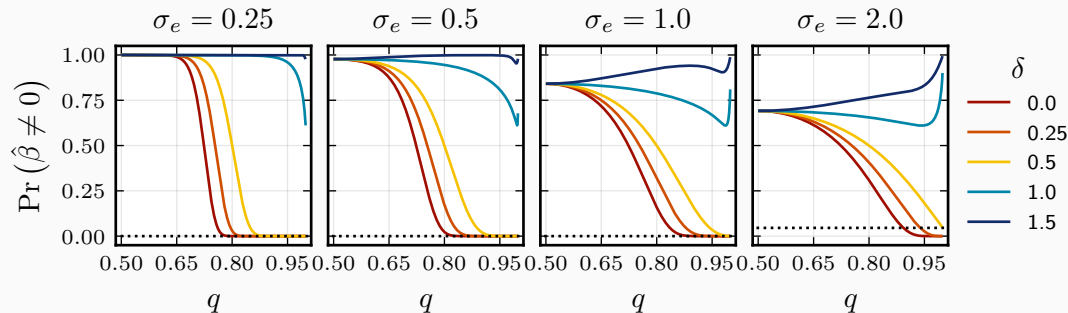
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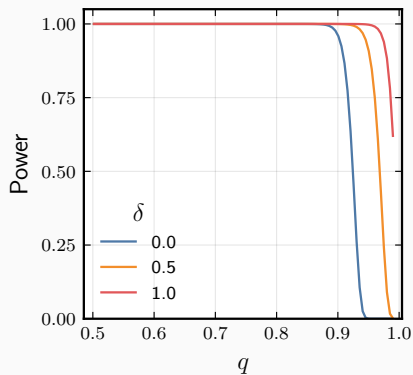
There is no (simple)  $s_j$  that will work for the elastic net.

## Probability of Selection

Since  $\mathbf{X}$  is fixed and  $\varepsilon$  is normal, we can compute the probability of selection.

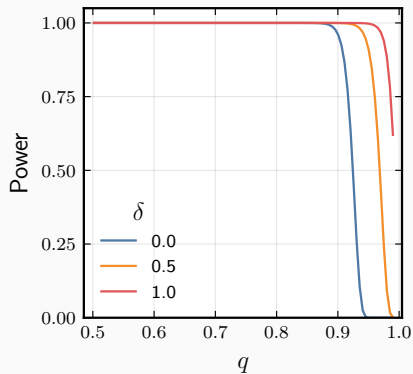


**Figure 7:** Probability that the elastic net selects a feature across different noise levels ( $\sigma_\varepsilon$ ), types of normalization ( $\delta$ ), and class balance ( $q$ ). The dashed line is asymptotic behavior for  $\delta = 1/2$ . Scaling used is  $s_j \propto (q - q^2)^\delta$ .

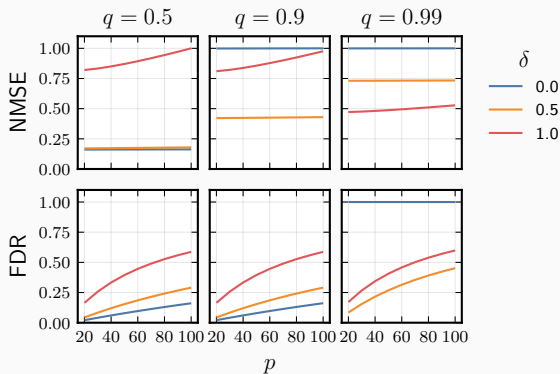


(a) Power in the sense of detecting all the true signals. Constant  $p$ .

**Figure 8:** Multiple features: 10 true signals and varying  $q$  and  $p$ . Mean squared error (MSE), false discovery rate (FDR), and power



(a) Power in the sense of detecting all the true signals. Constant  $p$ .



(b) False discovery rate (FDR) and normalized mean-squared error (NMSE).

**Figure 8:** Multiple features: 10 true signals and varying  $q$  and  $p$ . Mean squared error (MSE), false discovery rate (FDR), and power

## Rare Traits

Features with large class-imbalances might not be selected even if effect is **very strong** (e.g. rare SNPs, mutations).

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## Subgroup Data

Results become dependent on data collection.

Collecting more data with different class balances influences the results (since class balances change).

**So far:** all binary features. What about mixing binary and continuous (normal) features?

How to put binary features and normal features on the “same” scale?

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## One Definition of Comparability

The effects of a binary feature and a normally distributed feature are **comparable** if a flip in the binary feature has the same effect as a two-standard deviation change in the normal feature (Gelman 2008).

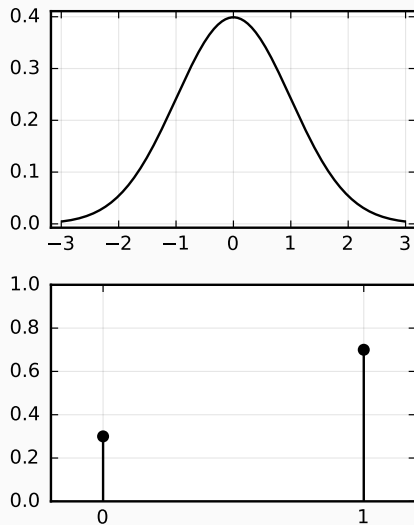
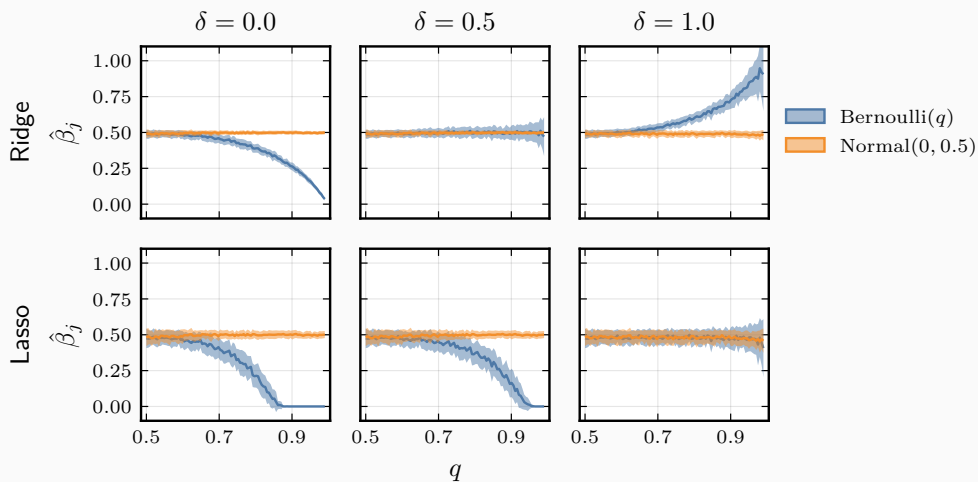


Figure 9: How do we match these?

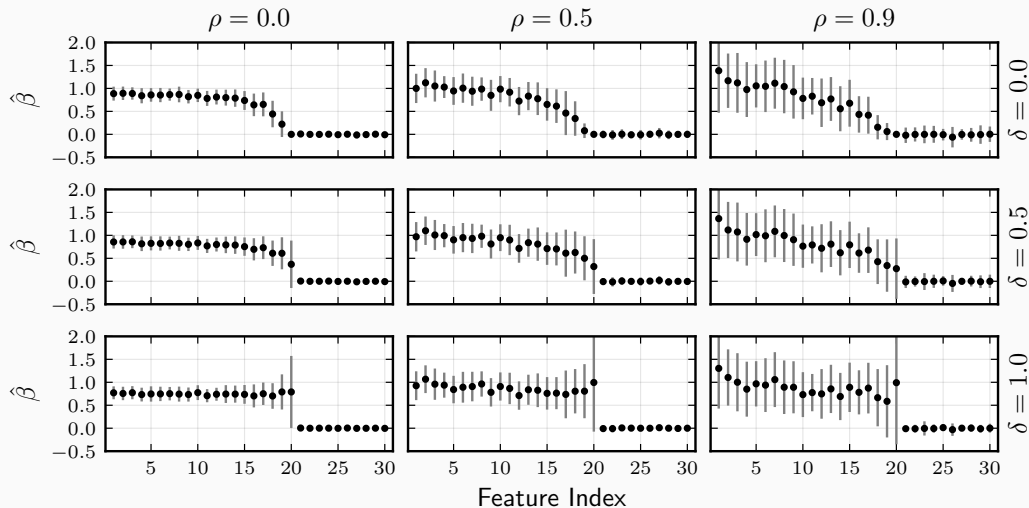


**Figure 10:** Comparison between lasso and ridge estimators for features generated to resemble features from various distributions.

## Experiments

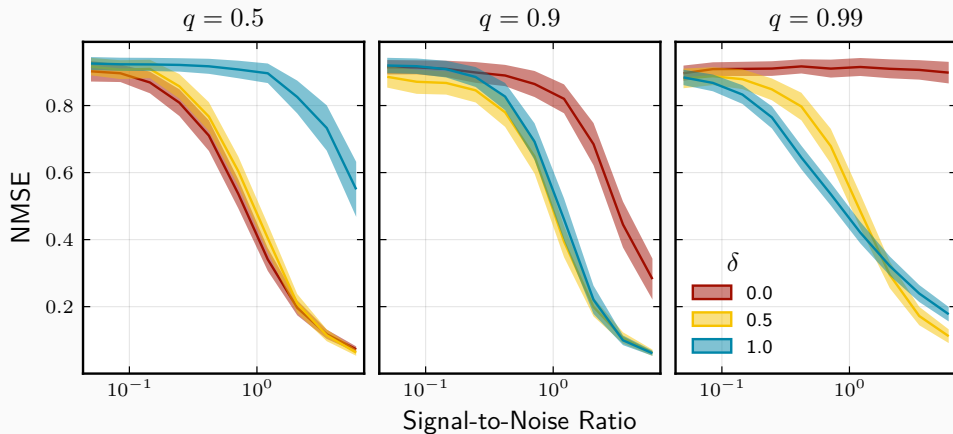
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## Binary Features (Decreasing $q$ )



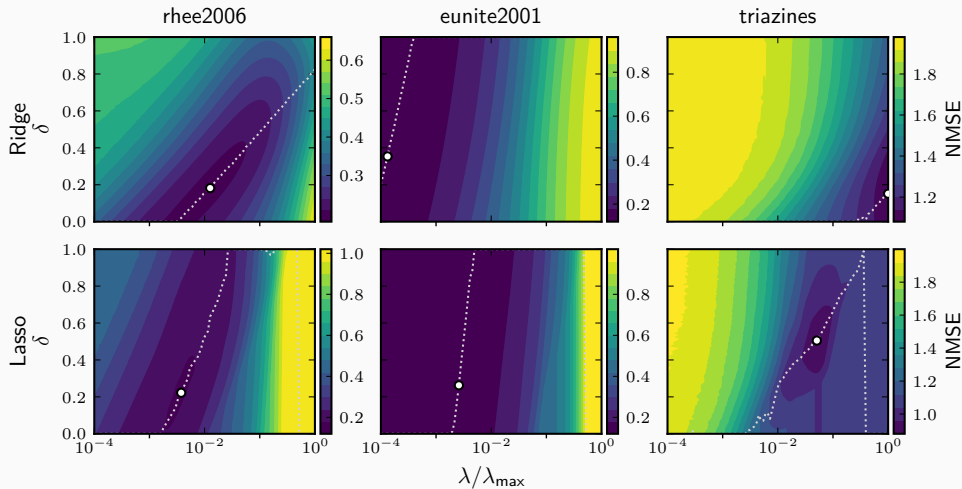
**Figure 11:** Lasso estimates for first 30 coefficients. First 20 features are true signals with a geometrically decreasing class balance from 0.5 to 0.99.  $\rho$  is a measure of autocorrelation.

## Binary Features (Signal-to-Noise Ratio)



**Figure 12:** Normalized mean-squared test set error (NMSE).

# Hyperparameter Optimization



**Figure 13:** Contour plots of hold-out (validation set) error across a grid of  $\delta$  and  $\lambda$  values for the lasso and ridge.

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## Additional Findings

- Intersections of binary/normal features create additional issues
- The only way to control the bias–variance tradeoff in the elastic net is to scale the penalty weights, not the features.

**Thank you!**

-  El Ghaoui, Laurent, Vivian Viallon, and Tarek Rabbani (Sept. 21, 2010). *Safe Feature Elimination in Sparse Supervised Learning*. Technical report UCB/EECS-2010-126. Berkeley: EECS Department, University of California. URL: <http://www.eecs.berkeley.edu/Pubs/TechRpts/2010/EECS-2010-126.html>.
-  Gelman, Andrew (July 10, 2008). “Scaling Regression Inputs by Dividing by Two Standard Deviations”. In: *Statistics in Medicine* 27.15, pp. 2865–2873. ISSN: 02776715, 10970258. DOI: [10.1002/sim.3107](https://doi.org/10.1002/sim.3107). URL: <https://onlinelibrary.wiley.com/doi/10.1002/sim.3107> (visited on 09/27/2023).
-  Larsson, Johan and Jonas Wallin (Nov. 2025). “The Choice of Normalization Influences Shrinkage in Regularized Regression”. In: *Transactions on Machine Learning Research*. ISSN: 2835-8856. (Visited on 12/05/2025).

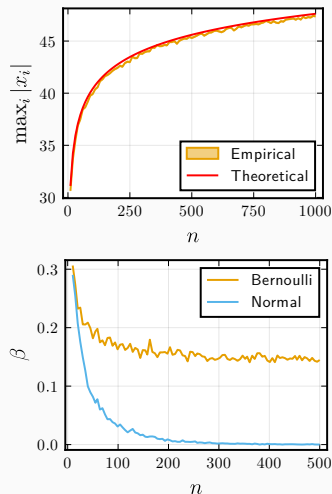
-  Tibshirani, Robert et al. (Mar. 2012). “Strong Rules for Discarding Predictors in Lasso-Type Problems”. In: *Journal of the Royal Statistical Society. Series B: Statistical Methodology* 74.2, pp. 245–266. ISSN: 1369-7412. DOI: [10/c4bb85](https://doi.org/10.1016/j.jrssb.2012.03.004). URL: <https://iths.pure.elsevier.com/en/publications/strong-rules-for-discarding-predictors-in-lasso-type-problems> (visited on 03/16/2018).
-  Zou, Hui and Trevor Hastie (2005). “Regularization and Variable Selection via the Elastic Net”. In: *Journal of the Royal Statistical Society. Series B (Statistical Methodology)* 67.2, pp. 301–320. ISSN: 1369-7412. URL: [www.jstor.org/stable/3647580](https://www.jstor.org/stable/3647580) (visited on 03/12/2018).

## Extras

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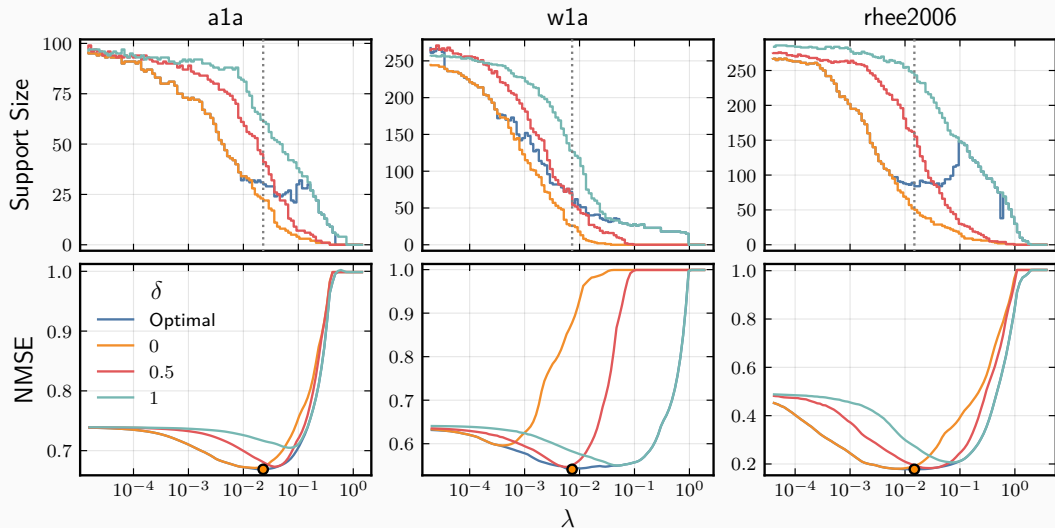
# Max-Abs Scaling of Continuous Features

- Min-max normalization is sometimes used in continuous data
- Very sensitive to outliers
- But also depend on sample size!
- In other words, results in model validation with varying sample sizes can yield very strange results.



**Figure 14:** Effects of maximum absolute value scaling.

# Hyperparameter Optimization (Support and NMSE)



**Figure 15:** Support and NMSE of the lasso for different values of  $\delta$  and  $\lambda$ .

- Proposed by Zou and Hastie ([2005](#)).

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- Lasso ( $\ell_1$ ) part:
  - Enables sparsity (interpretability, parsimony, feature selection)
  - Efficient when  $p \gg n$  (due to screening rules (El Ghaoui, Viallon, and Rabbani 2010; Tibshirani et al. 2012))

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- Ridge ( $\ell_2$ ) part
  - Mitigates lasso issue in correlated data
  - Better predictive performance when true signal is non-sparse